

Amin Saeidi Kelishami

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Profile

Computational biology and bioinformatics researcher with an AI background, focusing on multi-omics analysis, representation learning, and interpretable modeling for biological discovery.

Education

M.Sc. (M2) in Artificial Intelligence Sep 2025 – Present

Grenoble INP, Grenoble Alpes University, Grenoble, France

Thesis Title: Generative models for modeling microbial genomes

Supervisor: *Dr. Clovis Galiez*

Co-Supervisor: *Dr. Francois Enault*

M.Sc. in Computer Engineering (Specialized in Bioinformatics) Sep 2021 – Jun 2024

Sharif University of Technology, Tehran, Iran

GPA: 18.09/20

Thesis Title: Discovering Rational Properties of Chemicals by Rationalization and Causal Inference

Supervisor: *Dr. Babak Khalaj*

B.Sc. in Petroleum Engineering, Minor in Computer Science Sep 2016 – Feb 2021

Sharif University of Technology, Tehran, Iran

Thesis Title: Rate of Penetration Prediction and Optimization using AI methods

Supervisor: *Dr. Shahab Ayatollahi*

Co-Supervisor: *Dr. Ali Fahim*

Research Experience

Structural Variant Detection in Complex Genomic Regions, Jun 2025 – Nov 2025

- Integrated HiFi and Hi-C data to detect somatic structural variants in complex genomic regions (centromeres, telomeres).
- Developed methods for centromere annotation, including identification of monomers and higher-order repeats (HORs) in phased genomes to improve variant-calling resolution.

Single-Cell RNA-seq Analysis for Rare Disease, Jul 2025 – Sep 2025

- Analyzed single-cell RNA-seq datasets to characterize gene expression heterogeneity in rare disease samples.
- Integrated results with genomic variant data to identify potential biomarkers and disease-relevant cellular subpopulations.

Chemical Property Prediction with Interpretable GNNs, Jan 2023 – Jun 2024

- Developed an ensemble of GNN architectures with a generative explainer to identify molecular substructures driving chemical properties.
- Proposed statistical evaluation methods and interpretability metrics, validated through expert chemist review.

Teaching Experience

Teaching Assistant, Sharif University of Technology (Fall 2022 – Fall 2023)

- Courses: Computational Genomics, Machine Learning, Linear Algebra, Probability and Statistics

Mathematics & Physics Instructor, Private Schools (including Mofid High School), Sep 2016 – Jun 2019

Publications

- **Open Access**

- Amin Saeidi Kelishami - Mahmood Saeedi Kelishami - Sajjad Saeedi Kelishami, Cell Phone Image-Based Persian Rice Detection and Classification Using Deep Learning Techniques.

- **Conference Proceedings**

- Amin Saeidi Kelishami - Arash Imami Khiyavi - Ali Fahim - Shahab Ayatollahi, A Practical Approach with Ensemble-Driven Rate of Penetration Prediction and Optimization, International Conference on Technology and Energy Management (ICTEM 2024).
- Sara Bourbour Hosseinbeigi - Amin Saeidi Kelishami - Maryam Gheysari - Fatemeh Rahimzadeh, MetaDetox at TextDetox CLEF 2025: Detoxification with Few-Chain Prompting, Conference and Labs of the Evaluation Forum (CLEF 2025).
- Sara Bourbour Hosseinbeigi - Maryam Gheysari - Amin Saeidi Kelishami - Tahereh Talaei - Fatemeh Rahimzadeh - Erfan Moeini, TMU at SemEval-2025 Task 11: Tackling Multi-Label Emotion Detection with Transfer Learning, The 19th International Workshop on Semantic Evaluation (SemEval 2025).

Selected Academic Projects

Single-Cell RNA Sequencing Analysis, Spring 2023

- Used Python and Scanpy for preprocessing and clustering workflows.
- Preprocessing: QC, normalization, batch correction, and highly variable gene selection; clustering: marker gene identification and cluster annotation.

CellMembraneSegmentation - A MLOps Project, Spring 2023

- Preprocessed ISBI data, fine-tuned ResNet/VGG16 for classification, and developed U-Net for biomedical image segmentation in Keras.
- Used Docker for containerized deployment, MLflow for monitoring, and an automated CI/CD pipeline.

Bioinformatics Data Analysis, Fall 2021

- Analyzed microarray data for acute myeloid leukemia (AML) using R.
- Performed data quality control, dimension reduction, sample correlation, gene expression analysis, and gene ontology/pathway analysis.

Professional Experience

Bioinformatician, Pardis Gene, Sep 2024 – Sep 2025

- Conducted high-resolution HLA typing and variant calling for rare disease marker detection.
- Enhanced bioinformatics pipelines for accuracy and efficiency in variant analysis.
- Collaborated on genomic data interpretation and produced reports to support clinical decision-making.

NLP Team Lead, National Center for Artificial Intelligence, May 2024 – Sep 2024

- Led and mentored a team of NLP engineers across model development and delivery.
- Conducted code reviews, established evaluation metrics, and implemented best practices to improve model quality.

Natural Language Processing Specialist, National Center for Artificial Intelligence, Jan 2024 – May 2024

- Conducted NLP on social media text, images, and metadata, including sentiment analysis, named entity recognition, topic modeling, user profiling, and bot detection.
- Developed a Persian large language model and fine-tuned models including Meta Llama 3 for Persian performance improvements.

Knowledge Engineer, METAFAN Company, Jul 2023 – Jan 2024

- Developed knowledge representation methods to improve human-computer interaction.

Awards & Honors

- Awarded the **National Student-Entrepreneur Status (SNEE)** by Pepite oZer, Universite Grenoble Alpes, under the French Ministry of Higher Education.
- Ranked within the **top 0.2%** among more than 15,000 participants in the Iranian University Entrance Exam for Master's degree in Computer Engineering (May 2021).
- Ranked within the **top 0.3%** among more than 300,000 participants in the Iranian University Entrance Exam for Bachelor's degree (Aug 2016).

Skills

Programming Languages: Python, R, C/C++, Julia

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Keras, scikit-learn

Bioinformatics Tools & Pipelines: NGS data analysis, BLAST, Cytoscape, BEDTools, GATK, Samtools, BWA

Data Management: PostgreSQL, Elasticsearch

Developer Tools: Nextflow, Git, Docker, VS Code

Document Preparation: L^AT_EX

Operating Systems: Linux (WSL, server environments, HPCs)

Languages

Persian: Native

English: Fluent - **IELTS Academic:** *Overall: 7/9 (L: 7.5, R: 8, W: 6, S: 7)*

French: A1